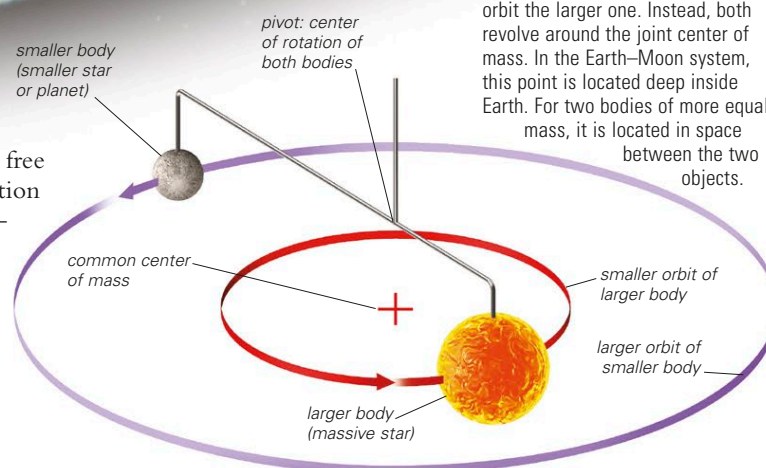
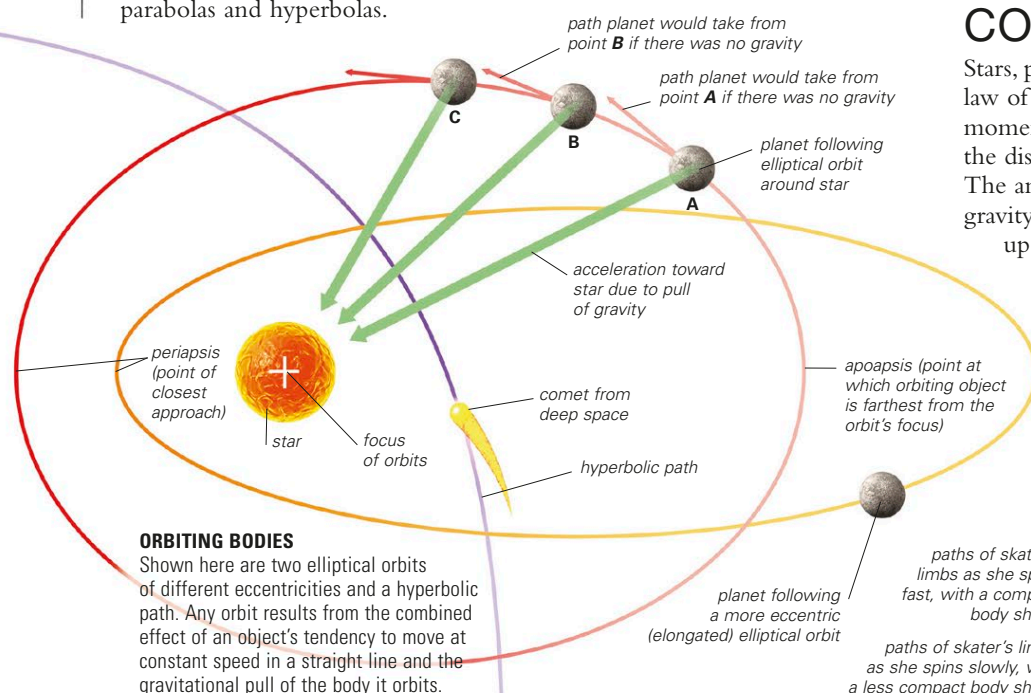


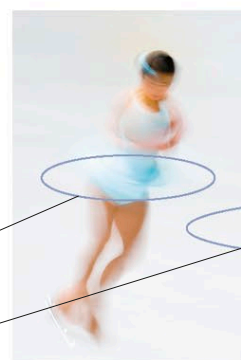
SHAPES OF ORBITS

When one object is in orbit around another object of higher mass, it is in free fall toward the larger body. It experiences a constant gravitational acceleration toward the larger object that deflects what would otherwise be its straight-line motion into a curved trajectory. The direction of its motion and the direction of acceleration both constantly change, producing its curved path. All closed orbits in nature have the shape of an ellipse (a stretched circle). The degree to which an ellipse varies from a perfect circle is called its eccentricity. Many solar system orbits (such as the Moon's around Earth) are not very eccentric—that is, they are almost circular. Others, such as Pluto's orbit around the Sun, are much more eccentric and highly elongated. Some celestial bodies follow open, nonreturning orbits along curves with shapes called parabolas and hyperbolas.



COMPACT, ROTATING BODIES

Stars, pulsars, galaxies, and planets all rotate, governed by the law of conservation of angular momentum. An object's angular momentum is related to its rotational energy, which depends on the distribution of mass in the object and on how fast it spins. The angular momentum of any rotating object stays constant, so if gravity causes the object to contract, its spin rate increases to make up for the redistribution of mass. Compact, fast-rotating objects therefore tend to form from diffuse, slowly rotating ones.



ANGULAR MOMENTUM

When an ice skater draws her limbs in, her spin rate soars. Similarly, a rotating cloud of gas spins faster as it contracts.

